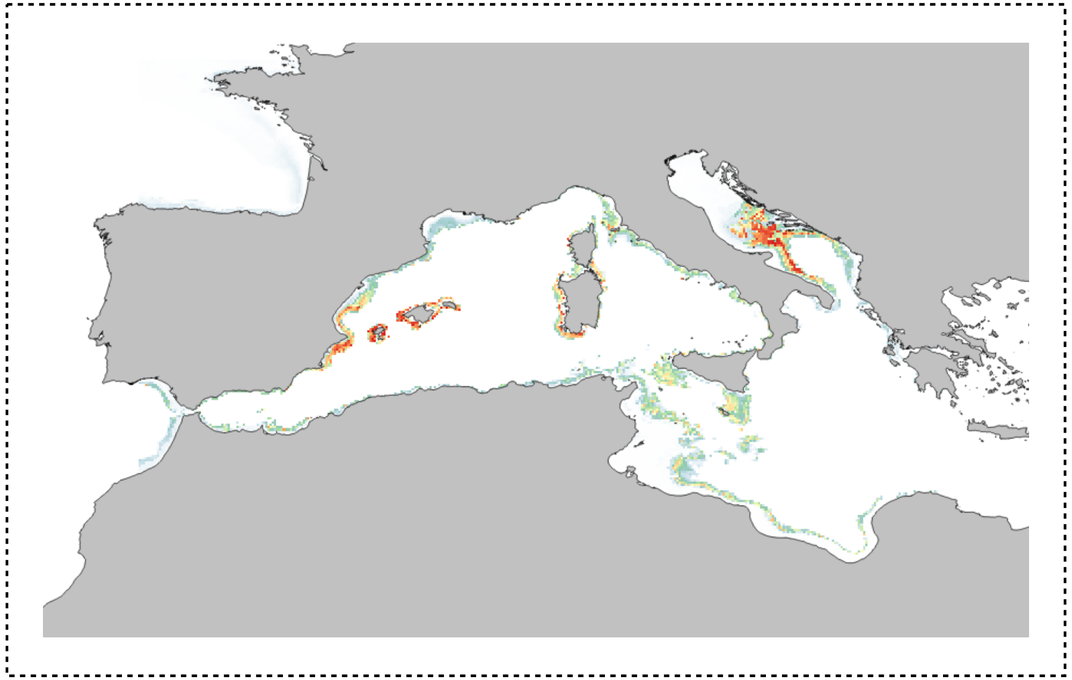
Evaluating, Transferring, and Ensembling

Modeling the distribution of biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Measuring SDM Performance



Model evaluation / performance

Verify consistency: do predictions match observations (presences and absences)?

Assess transferability: how well does the model generalize to new data?

Evaluate ecological realism: does the model make biological sense?

Fundamental question: *is the model suitable for the purpose (e.g., impact assessment)?*

Model evaluation

Comparing predictions with observations we assess for **‘false positives’** - model predicts occurrence where the species is absent - and **‘false negatives’** - model predicts absence where the species is present; **can be summarized in confusion matrices**.

Confusion matrix

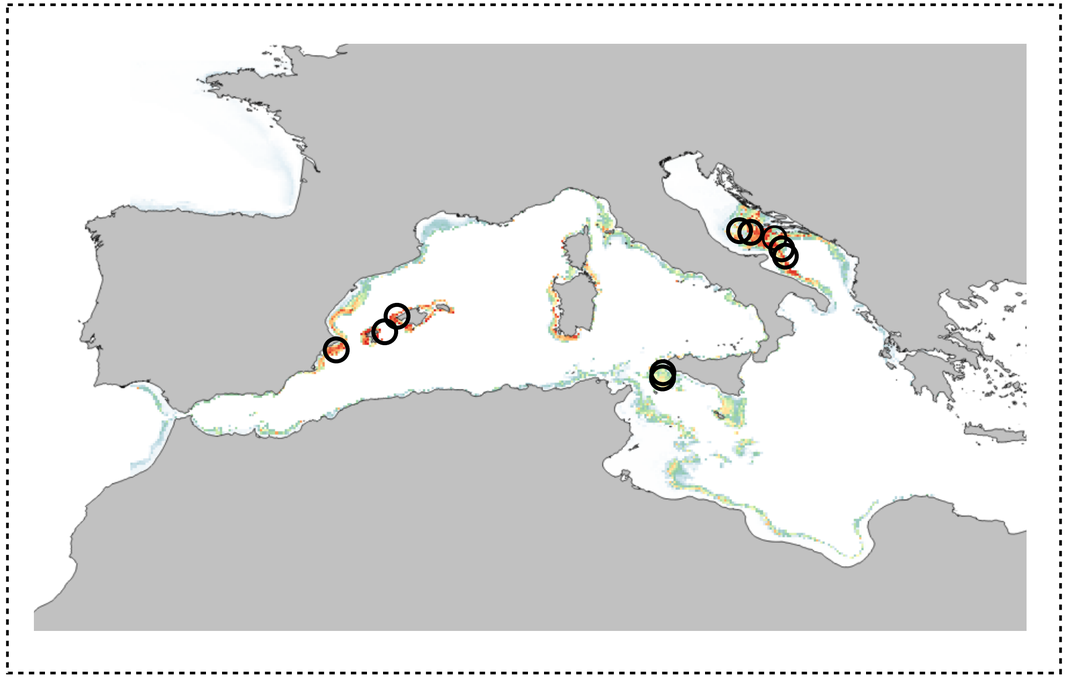
Types of prediction errors

		Observation	
		Presence	Absence
Prediction	Presence	True Positive	False Positive
	Absence	False Negative	True Negative

Perfect models only retrieve true positives and true negatives.

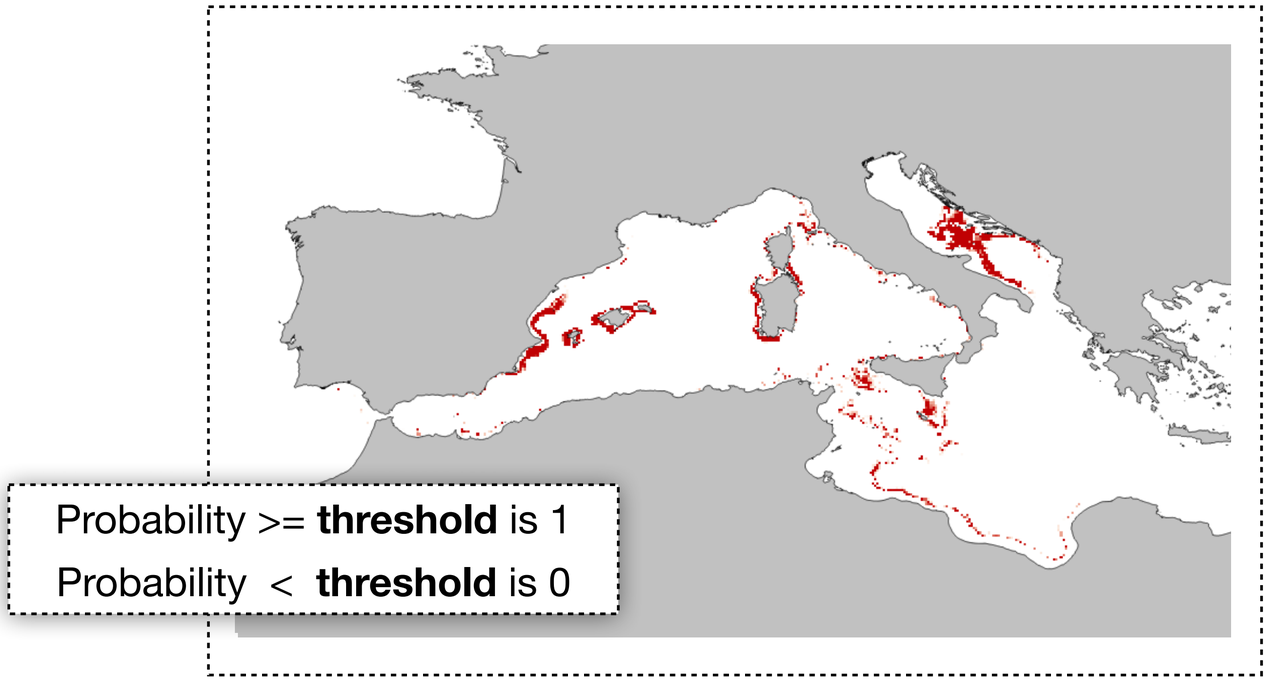
Predictions are continuous surfaces (e.g., probability, from 0 to 1).

One cannot compare observations (e.g., 1s) with the corresponding model output (e.g., P: 0.7).



Predictions are continuous surfaces (e.g., probability, from 0 to 1).

One cannot compare observations (e.g., 1s) with the corresponding model output (e.g., P: 0.7).



Predictions need to be reclassified as binary responses (presence-absence) for comparison with the observed data.

Evaluation metrics: threshold-dependent

After applying a threshold to predictions, the elements of the confusion matrix are used to compute various evaluation criteria that measure the performance of the model:

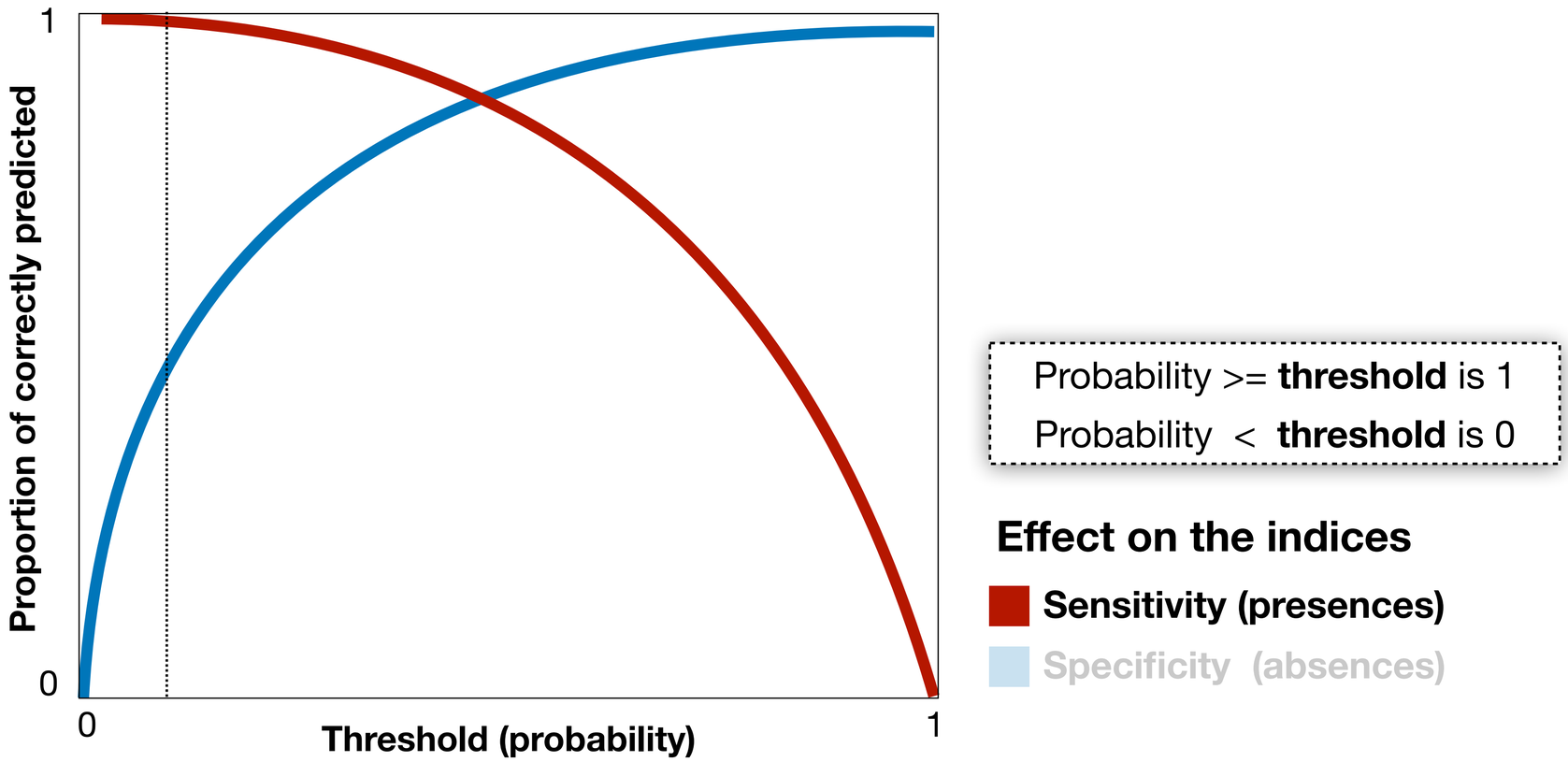
Sensitivity: proportion of presences correctly predicted, i.e., true positives;

Specificity: proportion of absences correctly predicted, i.e., true negatives;

True Skill Statistics: Sensitivity + Specificity - 1

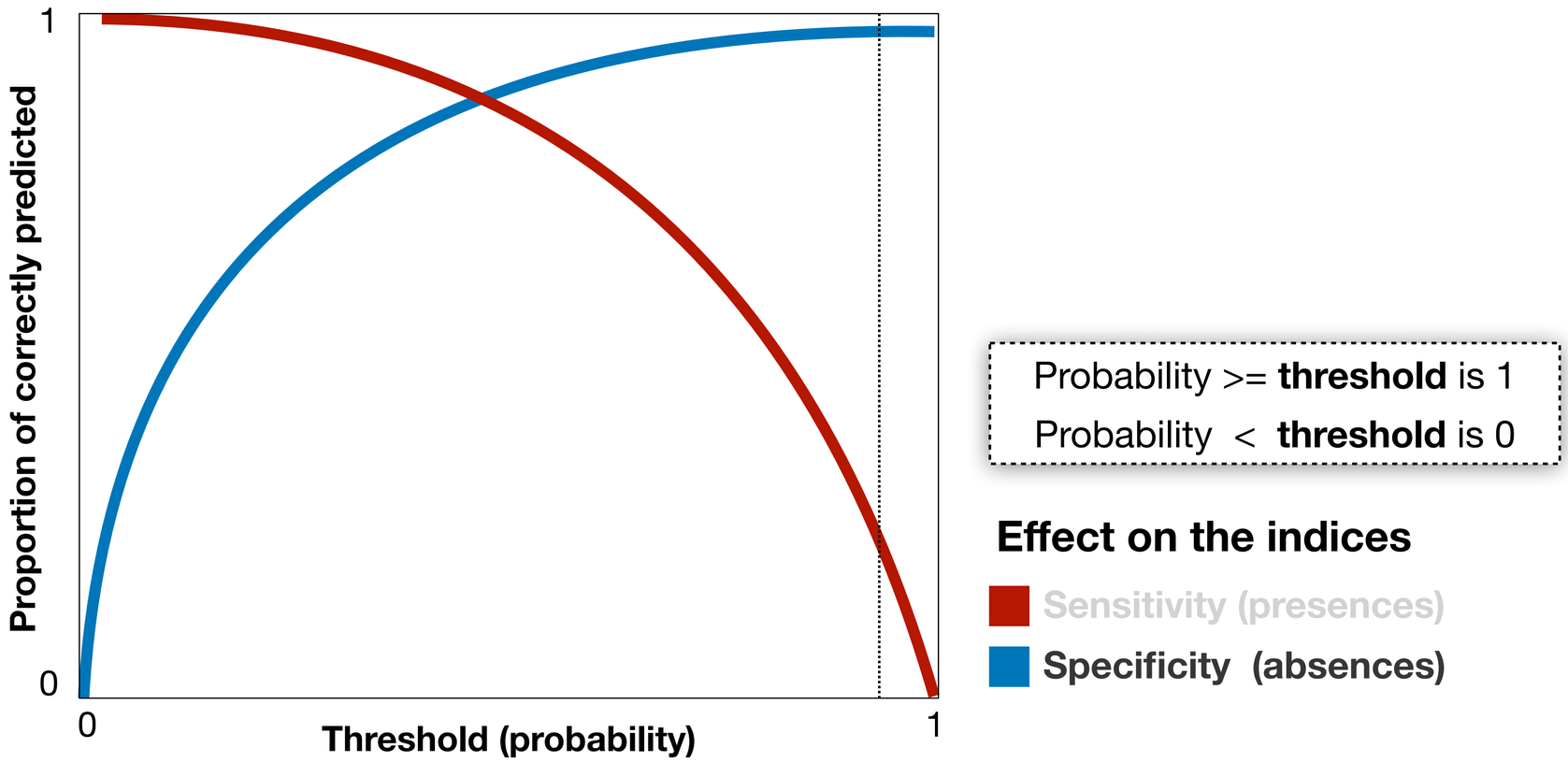
- TSS describes how well the model predicts presences and absences;
- TSS ranges from -1 to +1 (0 <= no better than a random model).

Which threshold to apply? Sensitivity and specificity vary with the range of thresholds used to reclassify predictions to binary outputs.



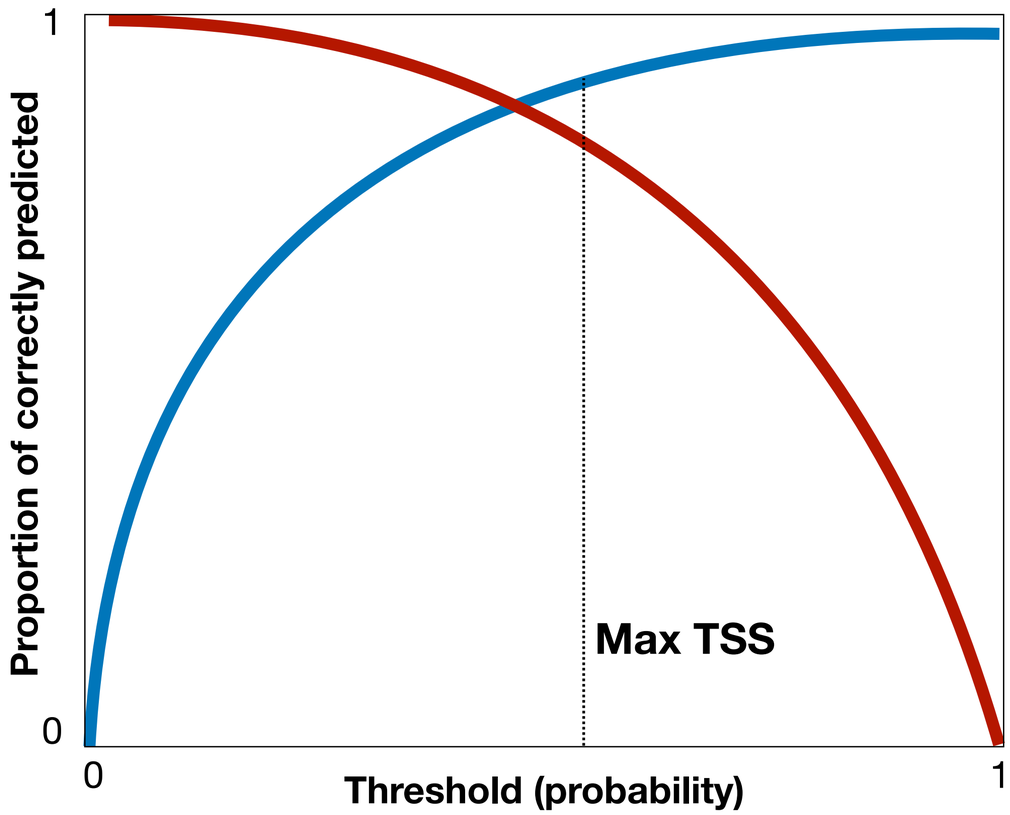
Using a low threshold most cells (in the map) will return 1.

Which threshold to apply? Sensitivity and specificity vary with the range of thresholds used to reclassify predictions to binary outputs.



Using a high threshold most cells (in the map) will return 0.

There are **threshold rules to maximize the agreement** between observed data and the predicted binary surface.



Probability >= **threshold** is 1

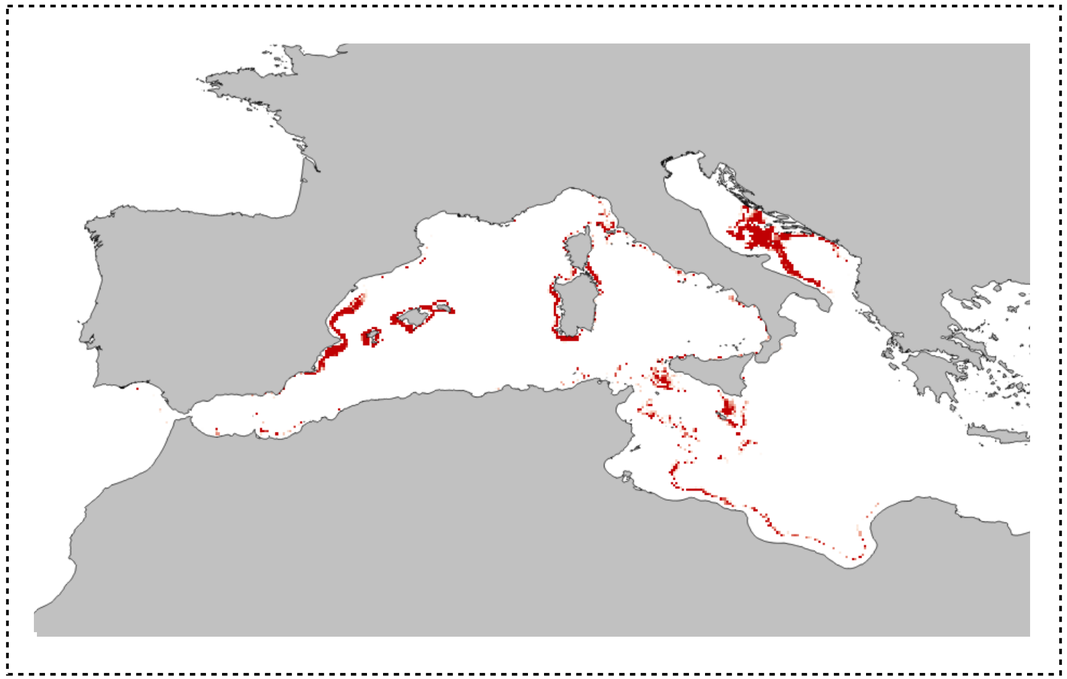
Probability < **threshold** is 0

Effect on the indices

Sensitivity (presences)

Specificity (absences)

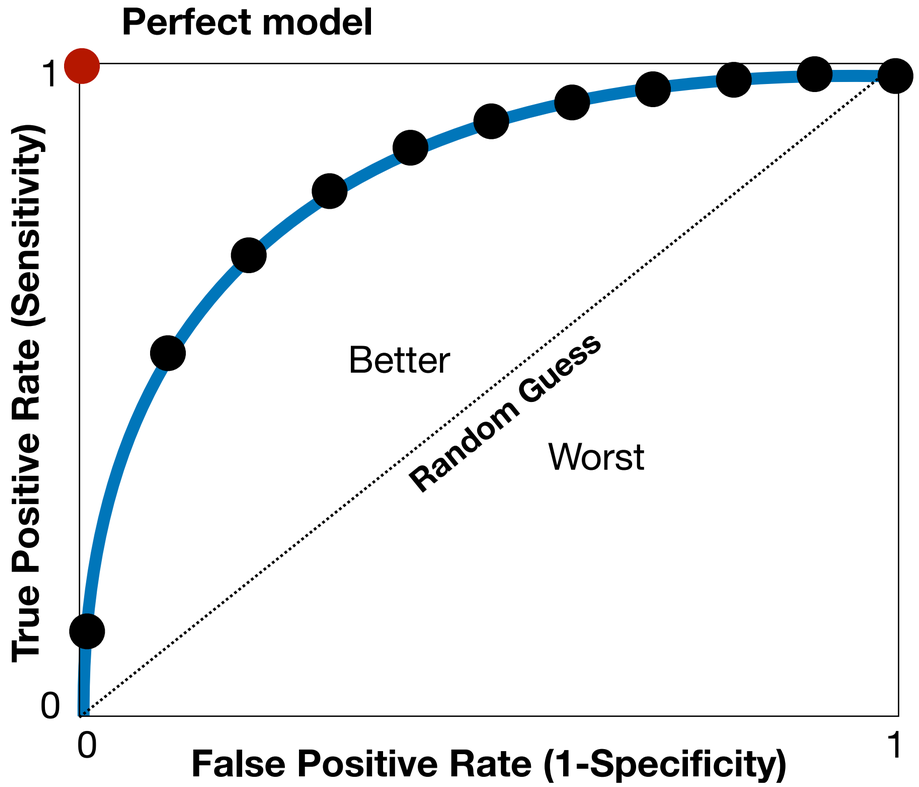
(!) **Maximization of sensitivity + specificity**



TSS = 0.93

Threshold-dependent measures of accuracy

Thresholds found using, e.g. “Max. TSS”, allow (1) **assessing the agreement between observed data and the model output** and (2) **converting probability maps into binary maps** - e.g., easier to discuss climate change impacts.



Threshold-independent evaluation criteria

Area **U**nder the **C**urve of the Receiver Operating Characteristic Plot.

Compare **true positive rate (sensitivity)** against **false positive rate (1-specificity)** across the range of all possible thresholds. The closer the curve from y-axis, the larger the **AUC**, and thus the more accurate the model.

AUC performance

Accuracy indices like AUC can be interpreted as follow:

1 - 0.9 : excellent model

0.9 - 0.8 : good model

0.8 - 0.7 : fair model

0.5 - 0.7 : poor model

0 - 0.5 : random model

The flaw of ROC-AUC in SDMs

It’s the most popular metric, but it can be highly misleading with imbalanced data, which is common in SDMs (e.g., 100 presences vs. 10,000 background points).

Why it Fails: It relies on specificity. Specificity is the true negative rate, so the score depends heavily on how well the model handles the negative class.

Easy background cells inflate it. Thousands of background cells far from the species’ conditions are trivially predicted as unsuitable. Each one adds a true negative and pushes ROC-AUC up.

The study area matters. The larger the extent, the more easy background cells, and the higher the ROC-AUC. The model doesn’t get any better.

To solve this, we use the **Precision-Recall Curve** and the **Boyce index**, which completely **ignore Specificity and focuses on Sensitivity**.

The Area Under the Precision-Recall Gain Curve (AUPRG): When the model predicts a presence, how often is it correct? Out of all presences, how many did the model successfully predicted?

Range 0 to 1 (0 = no better than a random model).

The Boyce index: Tests (by correlation) if higher modeled suitability corresponds to higher densities of presence points, which is the desired model behavior.

Range -1 to +1 (0 <= no better than a random model).

Model evaluation

Multi-faceted approach: don't rely on a single metric.

Use a suite:

Sensitivity;

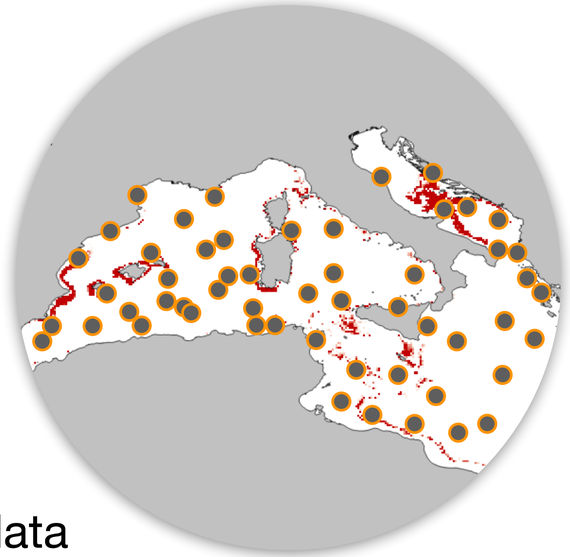
AUC-PRG;

Boyce;

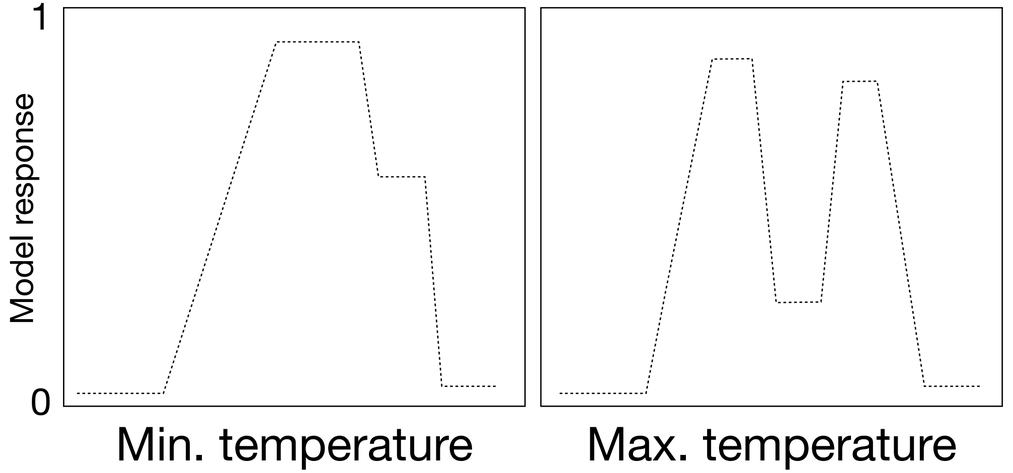
Rigorous Cross-Validation Strategies

High performance linked to good model transferability?

Depends on how it is measured.



● Testing data
○ Training data



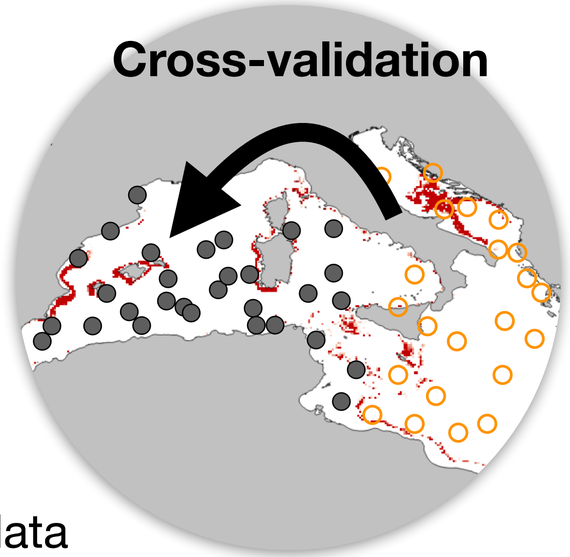
Testing performance with the training data leads to an overestimation of accuracy, regardless of the model's potential for transferability or if it ecologically sound.
If the models is largely overfitted - the test is performed in the locations of train - no way to assess for generality.

Huang et al., 2024

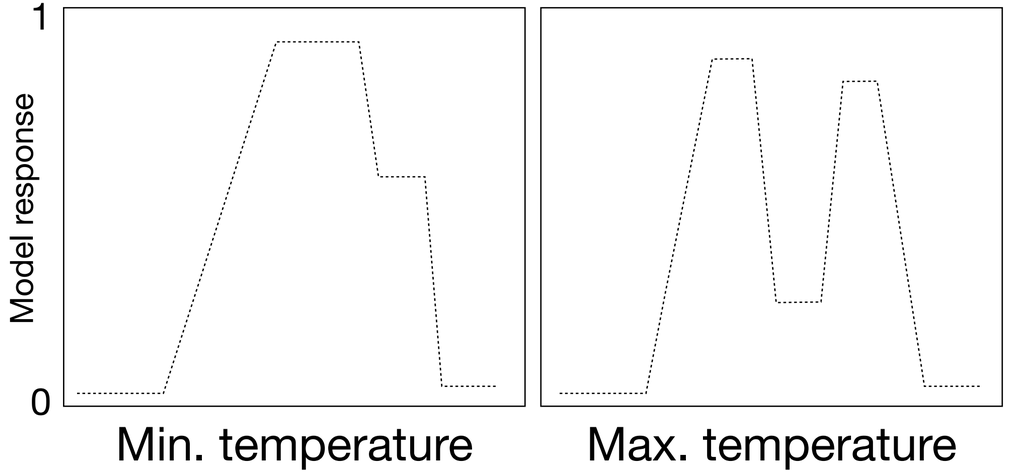
Rigorous Cross-Validation Strategies

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Testing accuracy with independent data is the approach to evaluate if the model generalized the relationship between occurrences and environment, and its transferability.

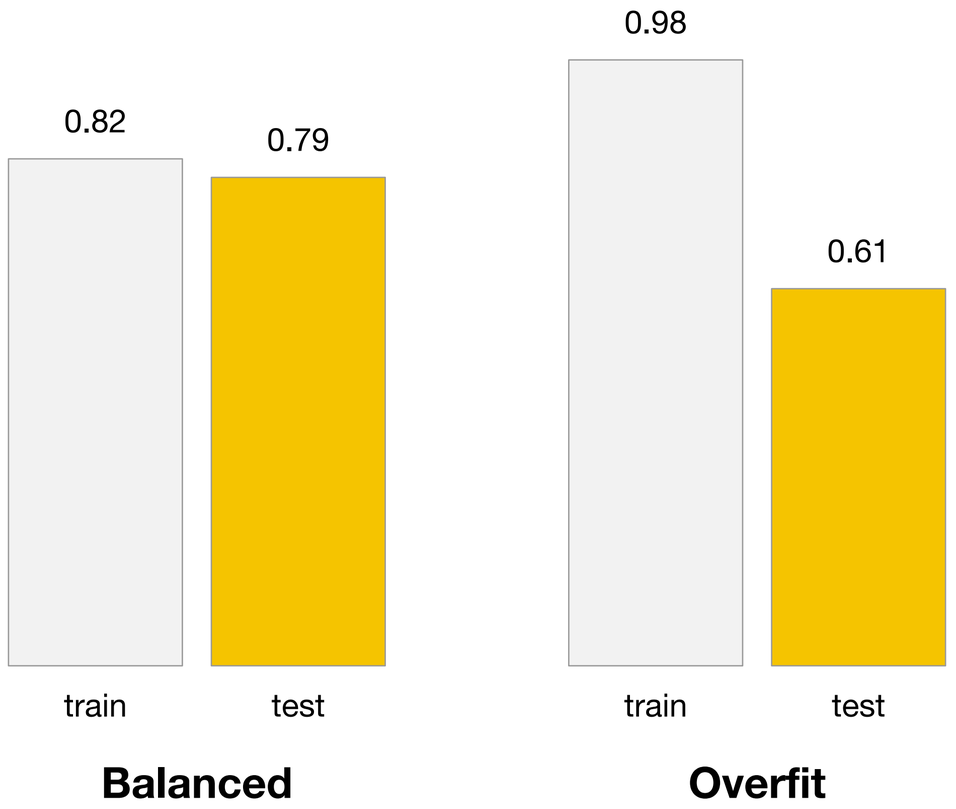
This is the same as projecting to other places or times.

Tends to lead to lower accuracy scores, but more reliable scores.

Huang et al., 2024

Detecting overfitting

Overfit models perform worse on validation or test data than on training data.



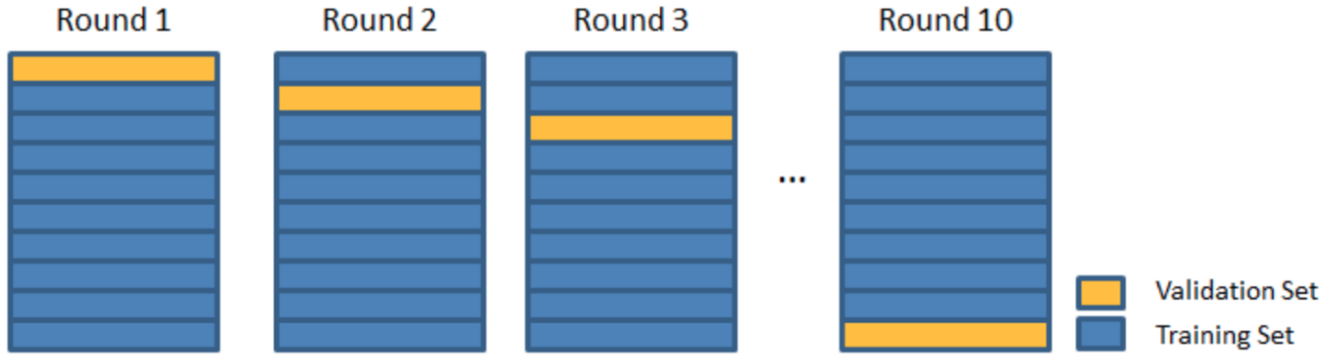
Missing independent data?

Problem: Niche truncation if occurrence data is yielded for testing.

Approach: partitioning the data in k cross-validation interactions, with data splits k times, yielding k estimates of accuracy that can be averaged.

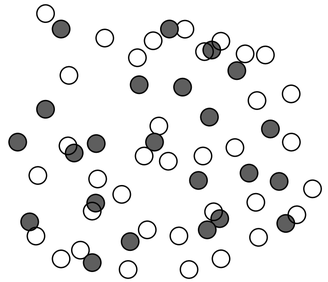
e.g.,

In 10-fold CV, 9/10 of the observations are **used to train the model** and the **remaining** 1/10 are **used to estimate performance**; this is repeated ten times and the estimated performance measures are averaged.



Missing independent data

There are different methods to produce independent datasets. **Some approaches provide more independent datasets than others**.



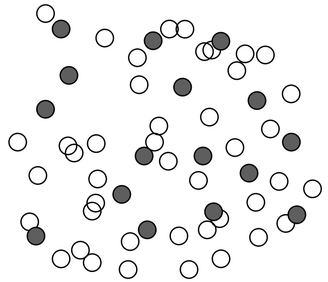
Random (70/30)

(70/30 | k-fold)

○ Training data ● Testing data

Missing independent data

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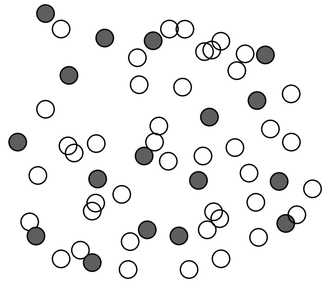
Random (70/30)

(70/30 | k-fold)

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Random (70/30)

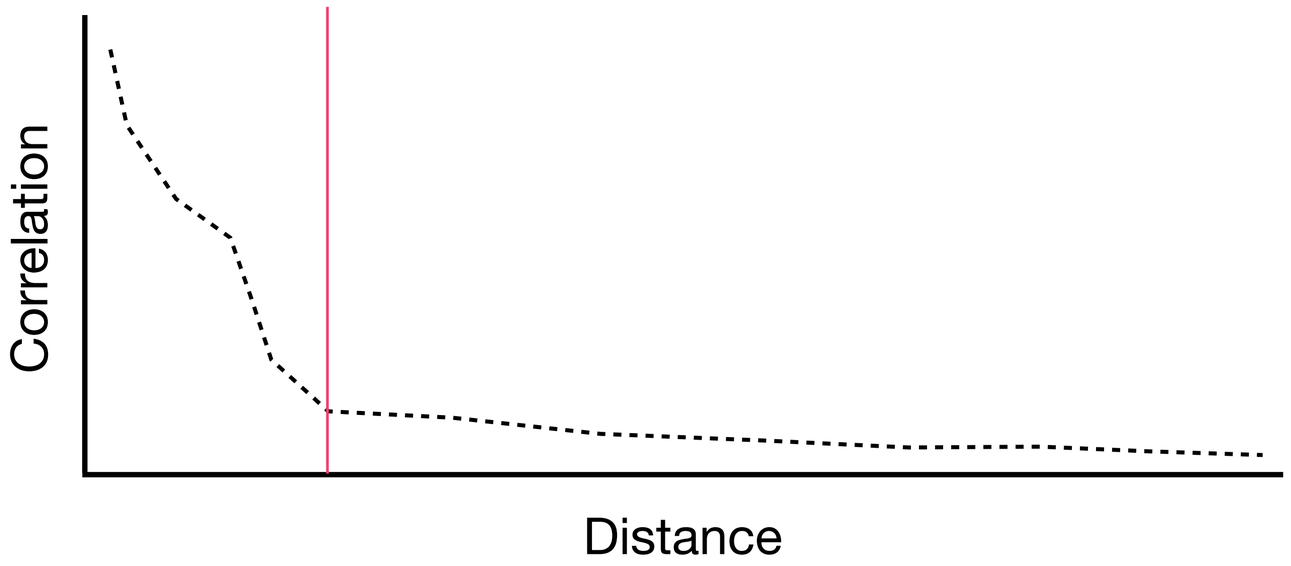
(70/30 | k-fold)

○ Training data ● Testing data

In standard (random) k-fold CV, train and test data are at nearby locations - often not independent.

Spatial autocorrelation: *”Near things are more related than distant things”.*

Can **overestimate performance** because training and testing folds contain similar data. **Violates the independence assumption** of statistical tests.



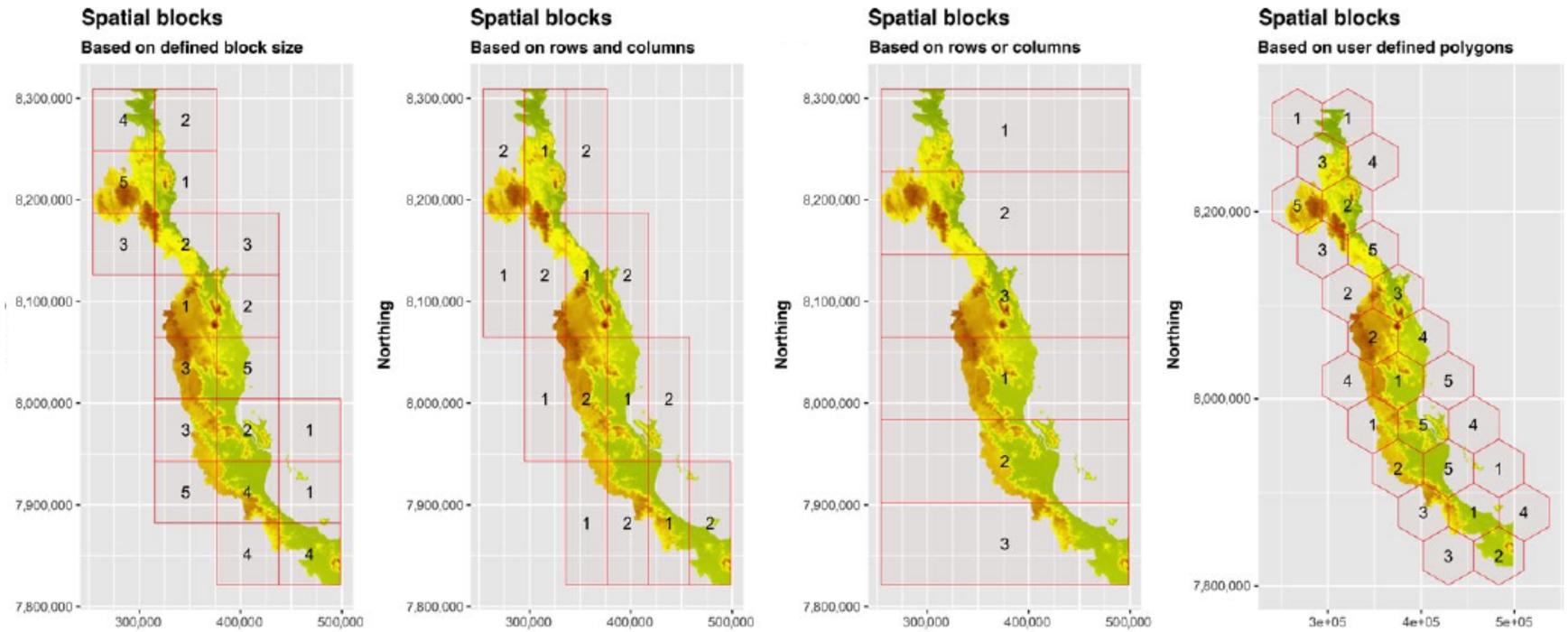
The consideration of Spatial Autocorrelation (SAC)

Records close to one another share nearly identical environmental conditions.

We estimate the distance at which records are not SAC - to define independence.

Missing independent data

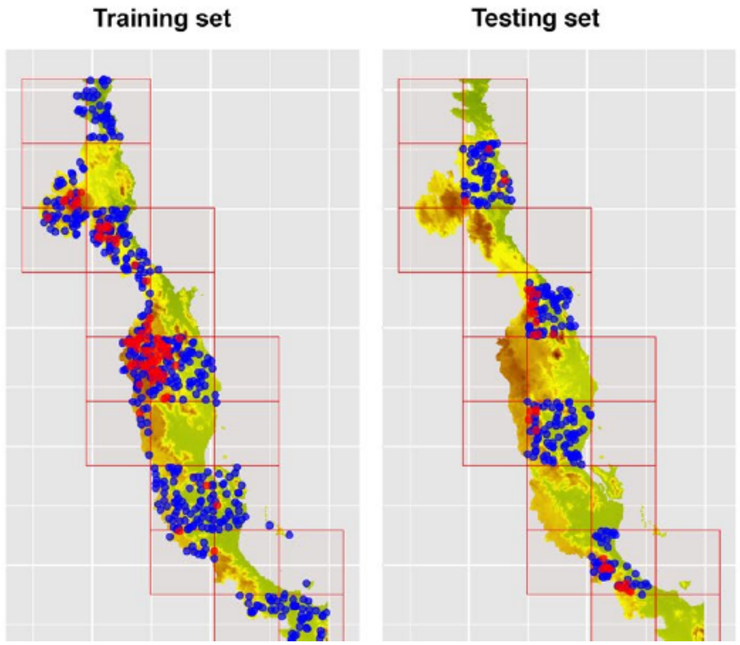
Spatial blocks considering SAC separate training and test data.
Verdict: block cross-validation is nearly universally more appropriate than random cross-validation.



Valavi et al., 2019

Missing independent data

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Verdict: block cross-validation is nearly universally more appropriate than random cross-validation.



Valavi et al., 2019

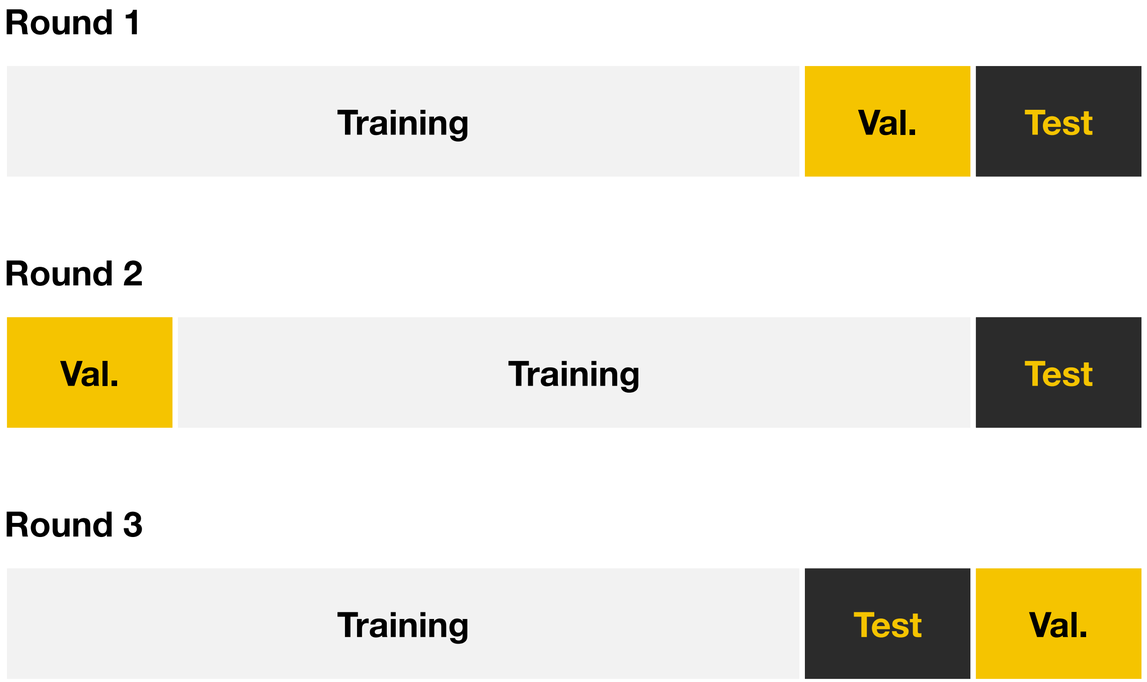
Training, validation and test

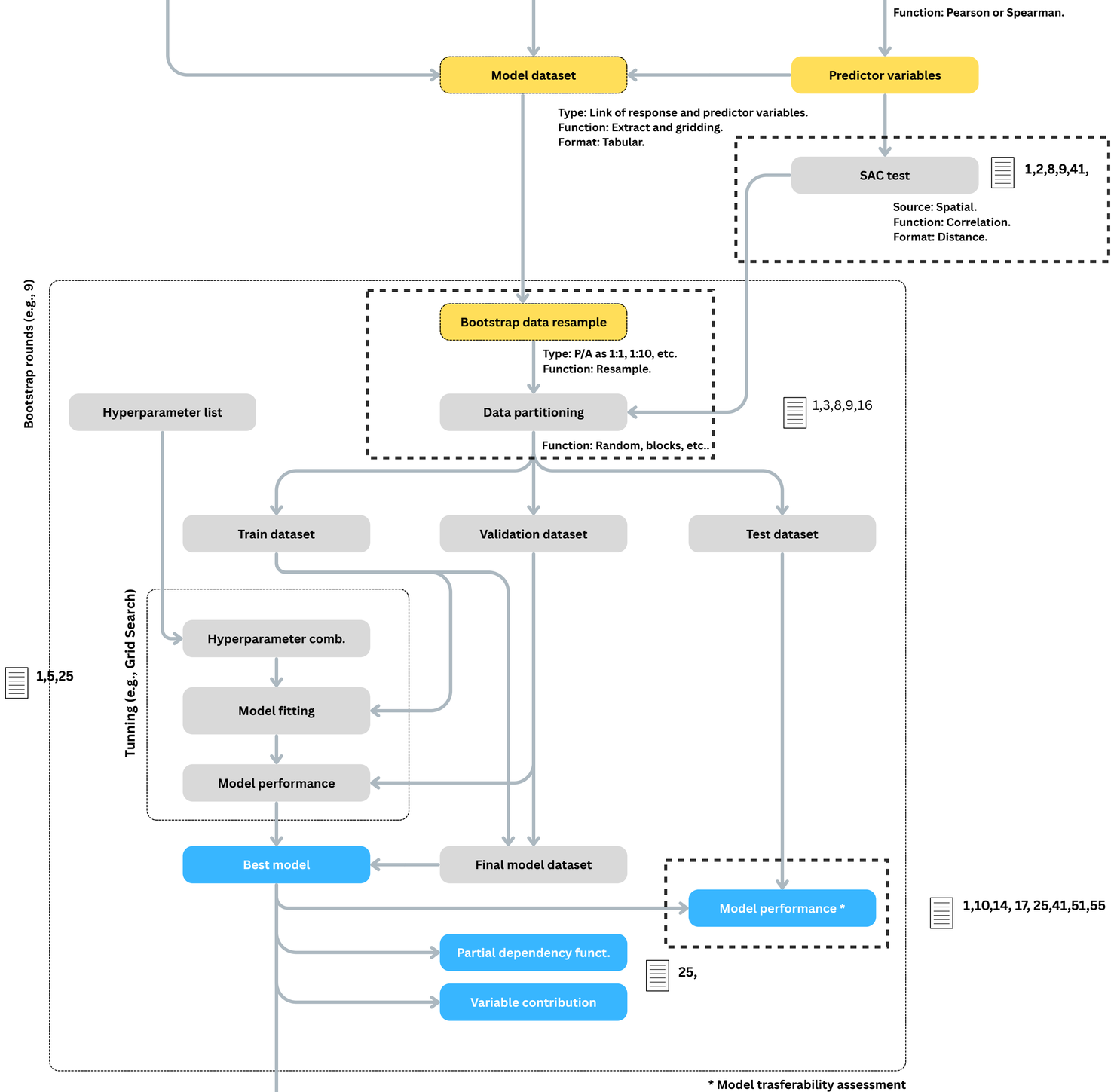
Training (70%): fits the model.

Validation (15%): selects the hyperparameters.

Test (15%): measures performance on data the model has not seen.

Repeated: in several rounds, with spatial blocks of the autocorrelation range from C09; the models of all rounds are averaged.




Function: Pearson or Spearman.
Model dataset
Predictor variables
Type: Link of response and predictor variables.
Function: Extract and gridding.
Format: Tabular.
SAC test
1,2,8,9,41,
Source: Spatial.
Function: Correlation.
Format: Distance.
Bootstrap rounds (e.g., 9)
Bootstrap data resample
Type: P/A as 1:1, 1:10, etc.
Function: Resample.
Hyperparameter list
Data partitioning
1,3,8,9,16
Function: Random, blocks, etc..
Train dataset
Validation dataset
Test dataset
Tunning (e.g., Grid Search)
Hyperparameter comb.
1,5,25
Model fitting
Model performance
Best model
Final model dataset
Model performance *
1,10,14, 17, 25,41,51,55
Partial dependency funct.
25,
Variable contribution
* Model trasferability assessment